

Table 5.1. Data stream package.

Data stream package	> S B1 B2 ... Bn <
Decimal view	62 83 B1 B2 ... Bn 60
Package length	27

Table 5.2. Data stream items.

Byte number	Item name	Item source	Item range (binary values)
B1	EMG CH1 low byte	DIAC0 ¹	0 – 255
B2	EMG CH1 high byte		0 – 3
B3	EMG CH2 low byte	DIAC1	0 – 255
B4	EMG CH2 high byte		0 – 3
B5	Hand strength (set-point) low byte	DIAC11	0 – 255
B6	Hand strength (set-point) high byte		0 – 3
B7	Hand battery status	DIAC13	0 – 100
B8	Main actual position	DIAC 75	0 – 255
B9	Hand mech grip state	DIAC 82	0 – lateral mode 1 – opposition mode 4 – neutral mode
B10	Hand motion type	DIAC 84	0 – lateral close 1 – opposition close 2 – opposition open 3 – lateral open
B11	Thumb force	DIAC 87	0 – 255
B12	Rotator actual position	DIAC 105	0 – 255
B13	Bit 1 (MSB) – Hand connection status Bit 2-8 - TR-FB battery status	TR-FB	0 – 255
B14 ²	Vibration motors statuses	TR-FB	0 – 255
B15	Vibration intensity of the motor 1	TR-FB	0 – 100
B16	Vibration intensity of the motor 2	TR-FB	0 – 100
B17	Vibration intensity of the motor 3	TR-FB	0 – 100
B18	Vibration intensity of the motor 4	TR-FB	0 – 100
B19	Vibration intensity of the motor 5	TR-FB	0 – 100
B20	Vibration intensity of the motor 6	TR-FB	0 – 100
B21	Vibration intensity of the motor 7	TR-FB	0 – 100
B22	Vibration intensity of the motor 8	TR-FB	0 – 100
B23	Hand speed TRFB-Control low byte	TR-FB	0 – 255

¹ DIAC - Data stream item activation code – according to Ottobock protocol (Michelangelo hand is the item source).

² B18 is calculated as follows:

$$B18 = \text{sum}([VB1, VB2, VB3, VB4, VB5, VB6, VB7, VB8] \cdot [2^0, 2^1, 2^2, 2^3, 2^4, 2^5, 2^6, 2^7]),$$

$$\text{where } VBn = \begin{cases} 0, & \text{if } VBn \text{ is turned off} \\ 1, & \text{if } VBn \text{ is turned on} \end{cases}$$

B24	Hand speed TRFB-Control high byte	TR-FB	0 – 3
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